MIL-A-3327A
27 December 1976
SUPERSEDING
MIL-S-3327
31 October 1950
Amendment-1
6 August 1959

MILITARY SPECIFICATION

ARCH SUPPORT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two classes, and three types of various sizes of non-metallic arch supports, furnished in pairs.

1.2 Classification.

1.2.1 Classes. The arch supports shall be of the following classes as specified:

Class A	-Female (Types I)
Class B	-Male (Types I, II and III)

1.2.2 Types. The arch supports shall be of the following types as specified:

Type I	-Combination (Longitudinal and Transverse)
Type II	-Metatarsal (Transverse)
Type III	-Plantar (Longitudinal)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Phila., Pa. 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

1.2.3 Sizes.

1.2.3.1 Class A arch supports shall be furnished in size 9, in B plus width.

1.2.3.2 Class B arch supports shall be furnished in sizes, 6, 7, 8, 9, 10, 11, and 12, all in D plus width.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-T-276	- Thread, Cotton.
KK-L-154	- Leather, Cattlehide, Bag and Case.
KK-L-271	- Leather; Cattlehide, Strap, Vegetable-Tanned.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Set-Up, Paperboard.

MILITARY

MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

FEDERAL

FED-STD-751	- Stitches, Seams and Stitching.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D1056: - Standard Specification for Sponge and Expanded Cellular Rubber Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

3. REQUIREMENTS

3.1 Material.

3.1.1 Arch cushions. The arch cushions shall be molded of open cell rubber. The cellular rubber shall be made from natural rubber, reclaimed rubber, or synthetic rubber, alone or in combination. The grade number shall be R015 in accordance with ASTM-D1056.

3.1.2 Stiffener. The stiffener shall be made of laminated phenolic impregnated fabric having a thickness of 1/32 inch to 3/64 inch, cut to the pattern of the insole in the case of type II supports, and molded to the heel in type I, and type III supports.

3.1.3 Adhesive. The adhesive shall be composed of crude or synthetic rubber together with pitch and resins dissolved in the proper grade of naptha. The adhesive shall contain not less than 0.4 pounds of rubber per gallon. The adhesive shall be heat resistant and shall be insoluble in water.

3.1.4 Leather.

3.1.4.1 Top covering pieces. The top covering pieces shall be fabricated of case leather conforming to type II of KK-L-154 or strap leather conforming to KK-L-271. The leather shall be mildew resistant, 3 ounces $\pm$ 1 ounce (2/64 inch to 4/64 inch thick). The color shall be natural as produced by the vegetable tanning process or brown. The leather shall otherwise be in accordance with KK-L-271, type III, class 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

3.1.4.2 Under leather pieces. The under leather pieces shall be fabricated of case leather conforming to KK-L-154, type II, class 1. The pieces shall be 3/64 inch to 6/64 inch in thickness. The finish shall be suede. The color shall be natural as produced by the vegetable tanning process.

3.1.5 Sewing thread. The sewing thread shall be type 1B3, ticket number 36, 4 ply, color brown, conforming to V-T-276. Colorfastness not applicable.

3.2 Dimension, shape, and contour.

3.2.1 Shape. The arch supports shall conform to the figures of this specification. Figures 1 to 21 inclusive, represents the outline shape of the right arch support for each size, type and class. For the left arch support, the outlines would be identical, except it shall have the reverse aspect.

3.2.2 Kleistone-Apex pattern numbers. The arch supports shall conform to the structural shapes of the Apex Foot Products Corp. patterns for the type and class (see 6.1) specified in table I.

TABLE I. Kleistone-Apex patterns.

Type	Class	Pattern No. (or equal)
I	A	200
I	B	700
II	B	825
III	B	110

3.2.3 Government Issue Shoes. The size, shape and contour of arch supports shall be suitable for use with regular Government Issue shoes.

3.3 Construction. The arch supports shall be constructed of a top covering of leather, a stiffener of impregnated fabric with cushions or pads as specified herein, and an underleather piece, all pieces shall be properly cemented together with first quality rubber cement and machined stitched along the edges through the top and bottom leather pieces only. The entire length and width of arch cushions and pads within the line of stitching shall be cemented to the leather.

3.3.1 Type I support. The type I support shall contain a rubber cushion molded in lefts and rights, extending the full width of the heel, within the line of stitching, tapering inward on the side edges as it extends to the front of the support. An impregnated fabric stiffener shall be set in the middle of the heel cushion and shall not cover the full heel. The metatarsal pad shall be placed forward of the rubber cushion. Top leather pieces shall be skived all around edge.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

3.3.2 Type II support. The type II support shall contain a heart shaped sponge rubber metatarsal pad cemented in metatarsal position molded in lefts and rights, and a stiffener of impregnated fabric shall be cemented the entire length and width of the cushion within the line of stitching. Top leather pieces shall be skived all around edge.

3.3.3 Type III support. The type III supports shall contain a molded cellular sponge rubber arch cushion molded in lefts and rights, and a stiffener of impregnated fabric cut to shape of the heel and cemented to the underpart of the sponge rubber at heel to prevent curling. The rubber arch cushion shall extend the full width of the heel, and then taper inward on the side edges as it extends to the front of the support. The impregnated fabric stiffiner shall be set in the middle of the heel cushion and shall not cover full heel.

3.4 Stitching. The stitching shall be not less than 1/32 inch or more than 1/8 inch from the edges provided such distances do not adversely affect the function of the arch supports. There shall be 12 to 22 threads per inch without causing leather chewing. There shall be a 1/2 inch overlap of stitches to serve as a reinforcement where stitches begin and end. The stitching shall conform to stitch type 301, and seam type SSa-1 in accordance with FED-STD-751.

3.5 Variations. It is anticipated that the makes of the different contractors may show variations, it is to be understood that the arch supports furnished shall conform in general with the dimensions and details herein specified to a degree consistent with the function for which they are intended.

3.7 Marking. Each support shall be marked with black indelible ink in block letters, 1/8 inch minimum height on the heel end of the under leather piece across the width showing the type and size number, the letter "F" or "M" to indicate female or male, "right heel" or "left heel" and the name or registered trademark of the contractor. (See figures)

3.8 Workmanship. Workmanship shall be first class throughout. The arch supports shall be free from defects which detract from their appearance or may impair their serviceability. Edges shall be clean cut.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in table II. The unit of product for examination purposes shall be one pair of arch supports of each type.

TABLE II. Sampling for examination.

Categories	Inspection level	AQL (percent defective)
Major	II	2.5
Minor	II	4.0

4.3 Examination.

4.3.1 Arch supports. The arch supports shall be examined for defects including, but not limited to, those listed in table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

TABLE III. Classification of defects.

Category	Defect
Major:	
101	Any portion of the upper surface rough or containing any size lump, void, blister, ripple, or any other defect that could cause the wearer discomfort.
102	Any portion twisted, puckered or pleated.
103	Stitching less than 1/32 inch or more than 1/8 inch from edge.
104	Any broken stitches.
105	Any portion of seam open.
106	Any hole, tear or cut in leather.
107	Any portion of stitching irregular or not gaged as specified.
108	Any needle chew.
109	Assembly not as specified.
110	Thread not type specified.
111	Seam not as specified.
112	Faulty adhesion, i.e.; top covering, stiffener, cushion and under leather piece separating.
113	Number of stitches per inch not as specified.
114	Stitching not overlapped as required.
115	Any deep cuts, damaged grain, open scratches, open grub holes, brands, or rough shoulders in leather.
Minor:	
201	Color not as specified.
202	Adhesive on outer surfaces.
203	Marking on arch support missing or illegible.
204	One or more spots or stains.
205	Any wrinkle and break patterns not affecting wearer comfort as compared to defect 101.

4.4 Packaging inspection. The inspection of the unit package, intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Tests. Tests shall be conducted in accordance with the applicable specifications, as necessary, to determine compliance with the specification.

5. PACKAGING

5.1 Packaging requirements. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Unit of issue. Two arch supports (one right, one left), as specified, constitutes one unit of issue, one pair.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

5.1.2 Level A.

5.1.2.1 Unit package. Each unit shall be packaged in a box constructed in accordance with PPP-B-566 or PPP-B-676.

5.1.2.2 Intermediate package. Twelve unit packages shall be packaged in a box constructed in accordance with PPP-B-566 or PPP-B-676.

5.1.3 Level C. Units shall be packaged in accordance with contractor's commercial practice which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Twelve intermediate packages (144 units) shall be packed in a box designed for a type 2 load, and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2, or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be in accordance with the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard boxes shall be waterproofed in accordance with PPP-B-636.

5.2.2 Level B. Twelve intermediate packages (144 units) shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The arch supports covered by this specification are intended for use in supporting the arch of the foot when worn with regular Government issue shoes.

6.2 Ordering data. Procurement documents should specify the following:

a) Title, number, and date of this specification.
b) Class, size and type required (see 1.2.1, 1.2.2 and 1.2.3).
c) National stock number (NSN).
d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification is intended to cover the following items appearing in the Federal Supply Catalog:

National stock number	Item identification
6515-00-299-8004	ARCH SUPPORT, PLANTAR, MALE SIZE 8
6515-00-299-8005	ARCH SUPPORT, PLANTAR, MALE SIZE 11
6515-00-299-8009	ARCH SUPPORT, PLANTAR, MALE SIZE 12
6515-00-299-8010	ARCH SUPPORT, METATARSAL, MALE SIZE 12
6515-00-299-8012	ARCH SUPPORT, METATARSAL, MALE SIZE 10
6515-00-299-8026	ARCH SUPPORT, PLANTAR, MALE SIZE 9
6515-00-299-8027	ARCH SUPPORT, PLANTAR, MALE SIZE 10
6515-00-299-8028	ARCH SUPPORT, PLANTAR, MALE SIZE 6
6515-00-299-8029	ARCH SUPPORT, PLANTAR, MALE SIZE 7
6515-00-299-8040	ARCH SUPPORT, COMBINATION, FEMALE SIZE 9
6515-00-299-8060	ARCH SUPPORT, COMBINATION, MALE SIZE 6
6515-00-299-8061	ARCH SUPPORT, COMBINATION, MALE SIZE 7
6515-00-299-8062	ARCH SUPPORT, COMBINATION, MALE SIZE 10
6515-00-299-8063	ARCH SUPPORT, COMBINATION, MALE SIZE 11
6515-00-299-8064	ARCH SUPPORT, COMBINATION, MALE SIZE 8
6515-00-299-8065	ARCH SUPPORT, COMBINATION, MALE SIZE 9
6515-00-299-8066	ARCH SUPPORT, COMBINATION, MALE SIZE 12
6515-00-299-8071	ARCH SUPPORT, METATARSAL, MALE SIZE 11
6515-00-299-8072	ARCH SUPPORT, METATARSAL, MALE SIZE 8
6515-00-299-8073	ARCH SUPPORT, METATARSAL, MALE SIZE 9
6515-00-299-8075	ARCH SUPPORT, METATARSAL, MALE SIZE 7

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Project No. 6515-2055

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

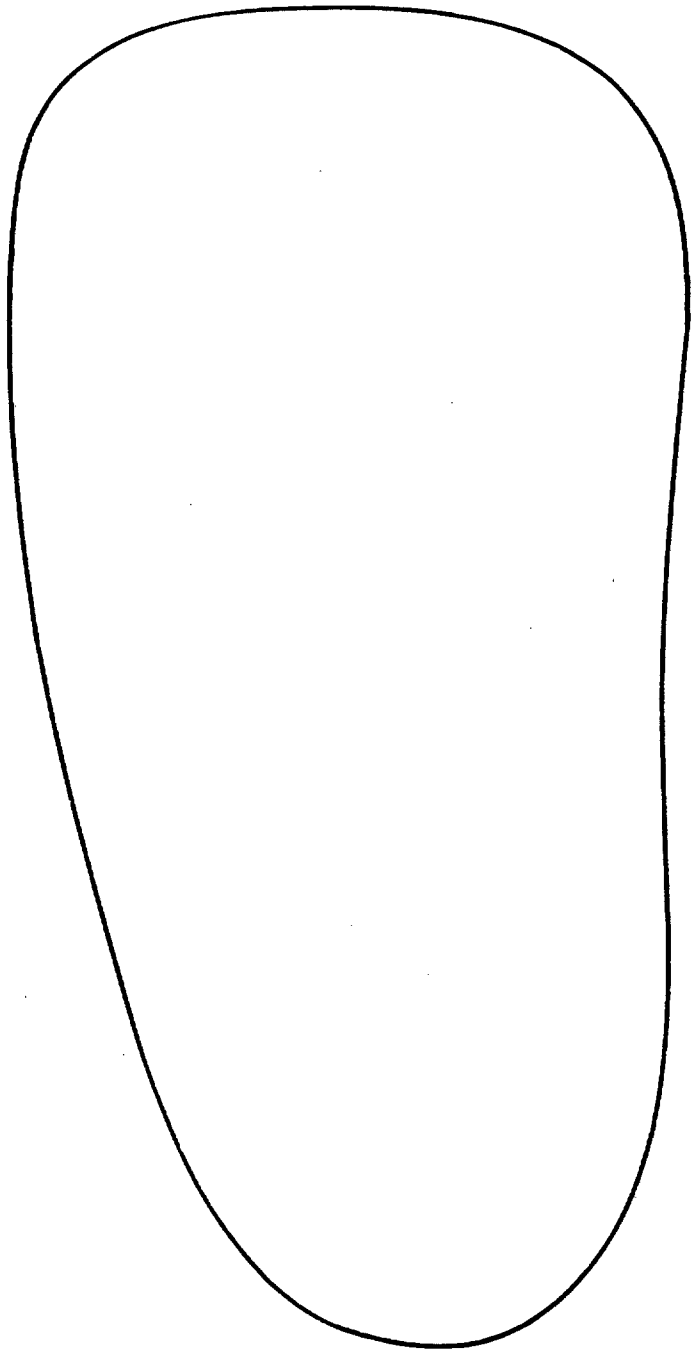


FIGURE 1. Arch support, plantar, male size 8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

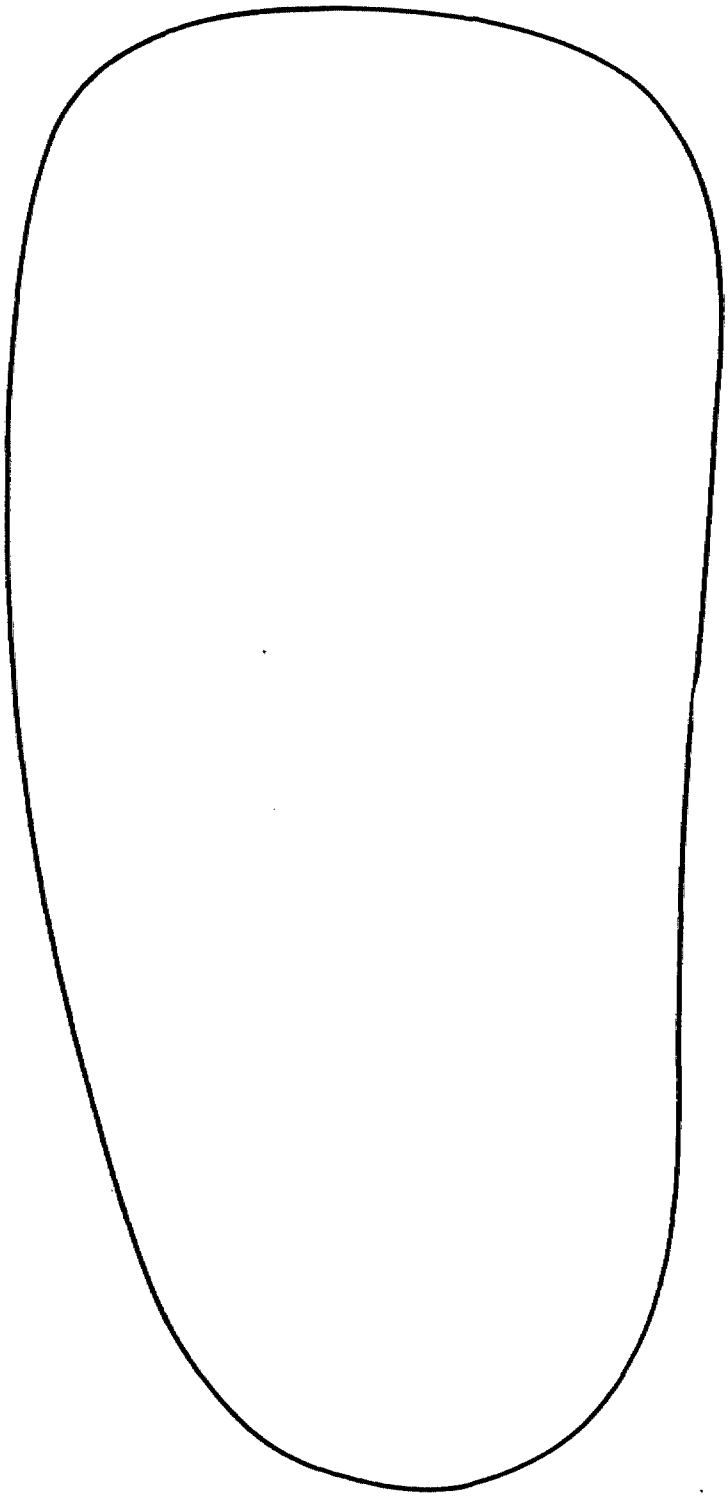


FIGURE 2. Arch support, plantar, male size 11.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

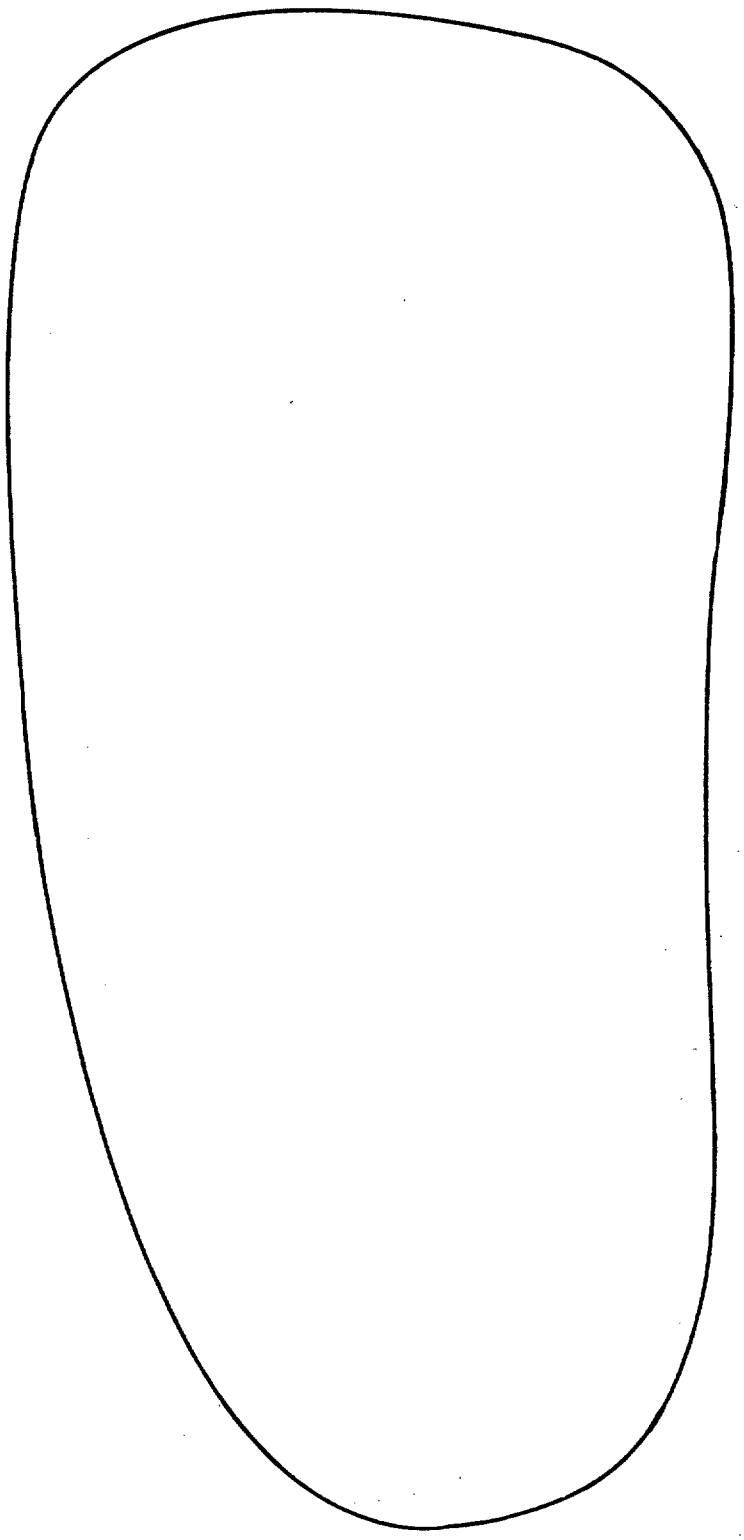


FIGURE 3. Arch support, plantar, male size 12.

Source: https://assist.dla.mil Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

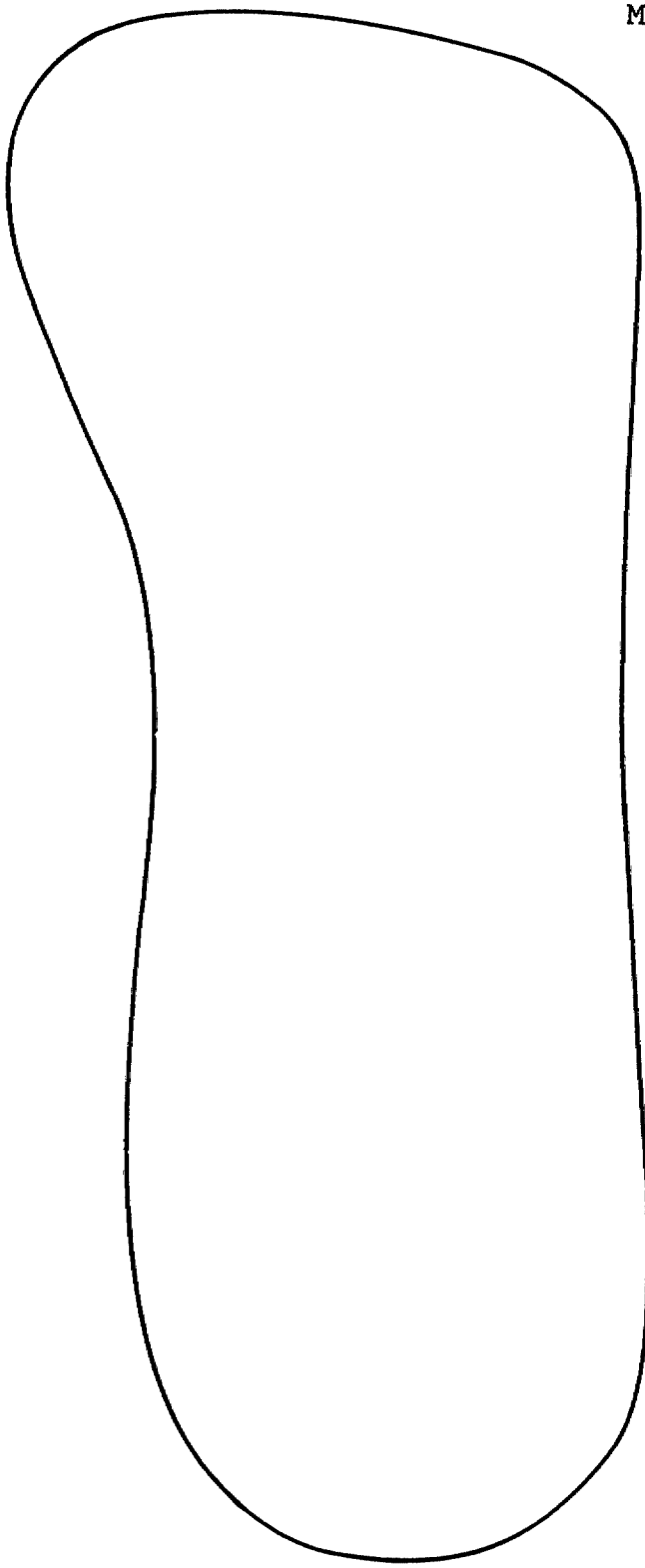


FIGURE 4. Arch support, metatarsal, male size 12.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

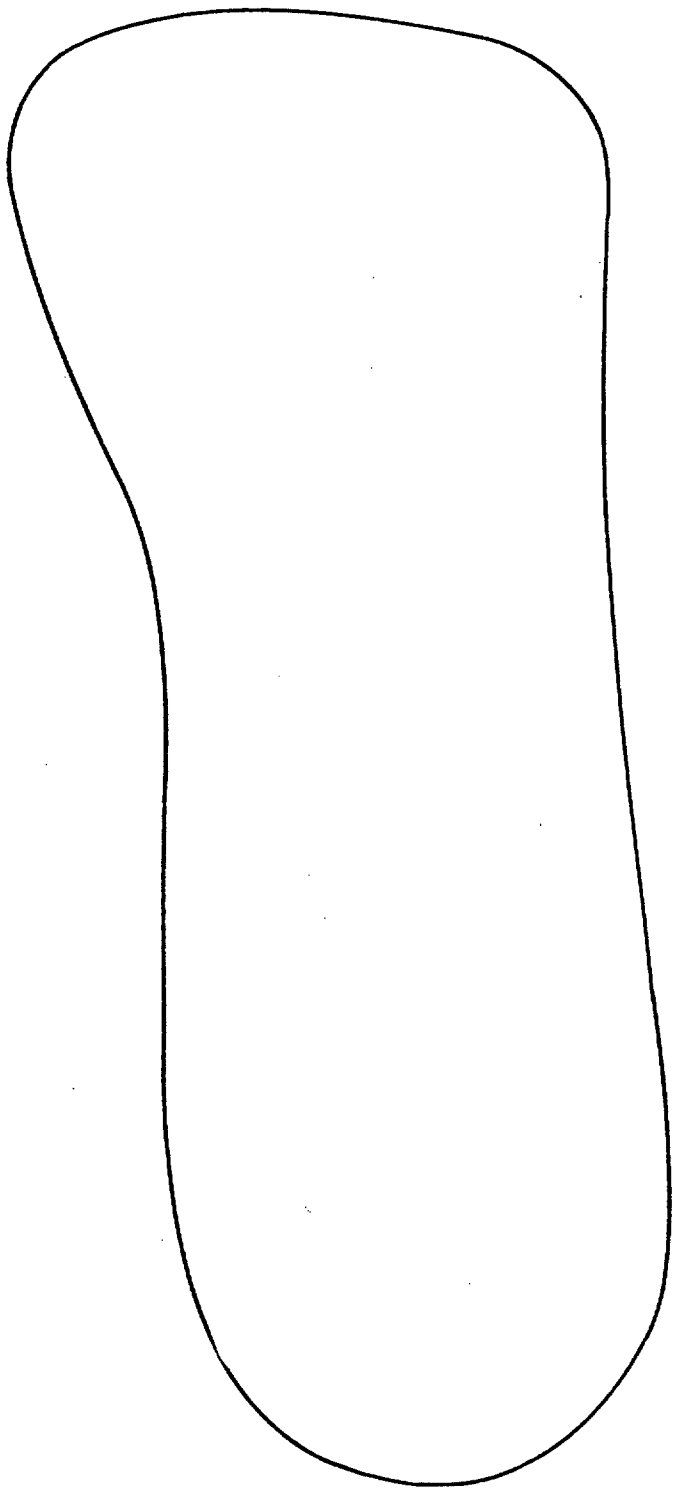


FIGURE 5. Arch support, Metatarsal, male size 10.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

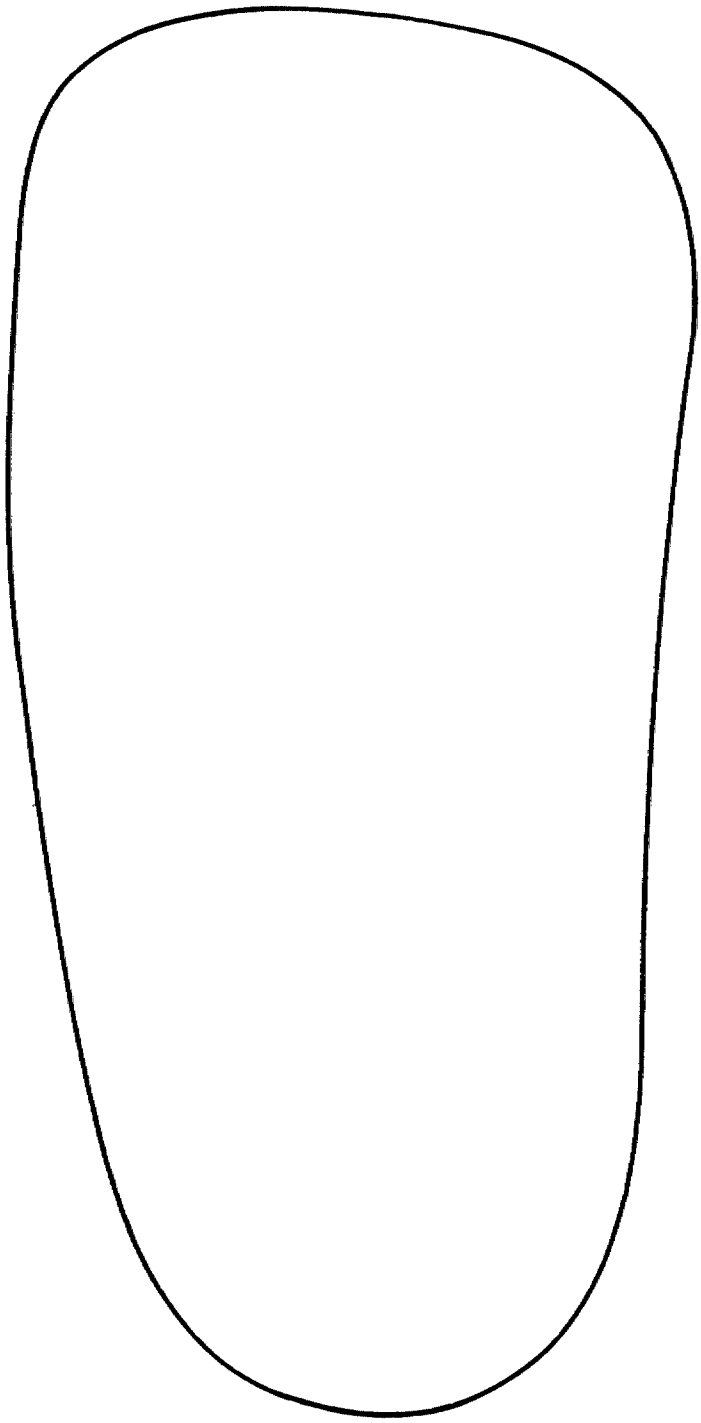


FIGURE 6. Arch support, plantar, male size 9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

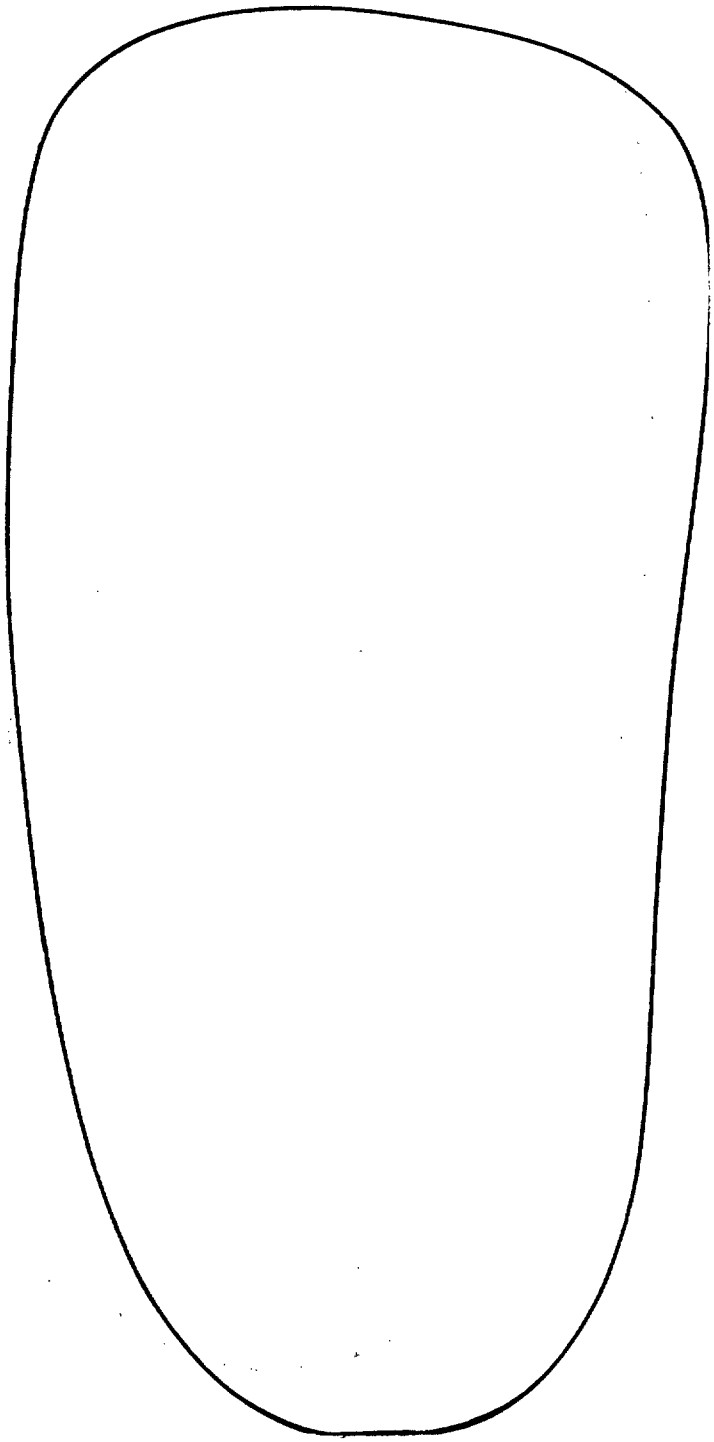


FIGURE 7. Arch support, plantar, male size 10.

Source: https://assist.dla.mil --Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

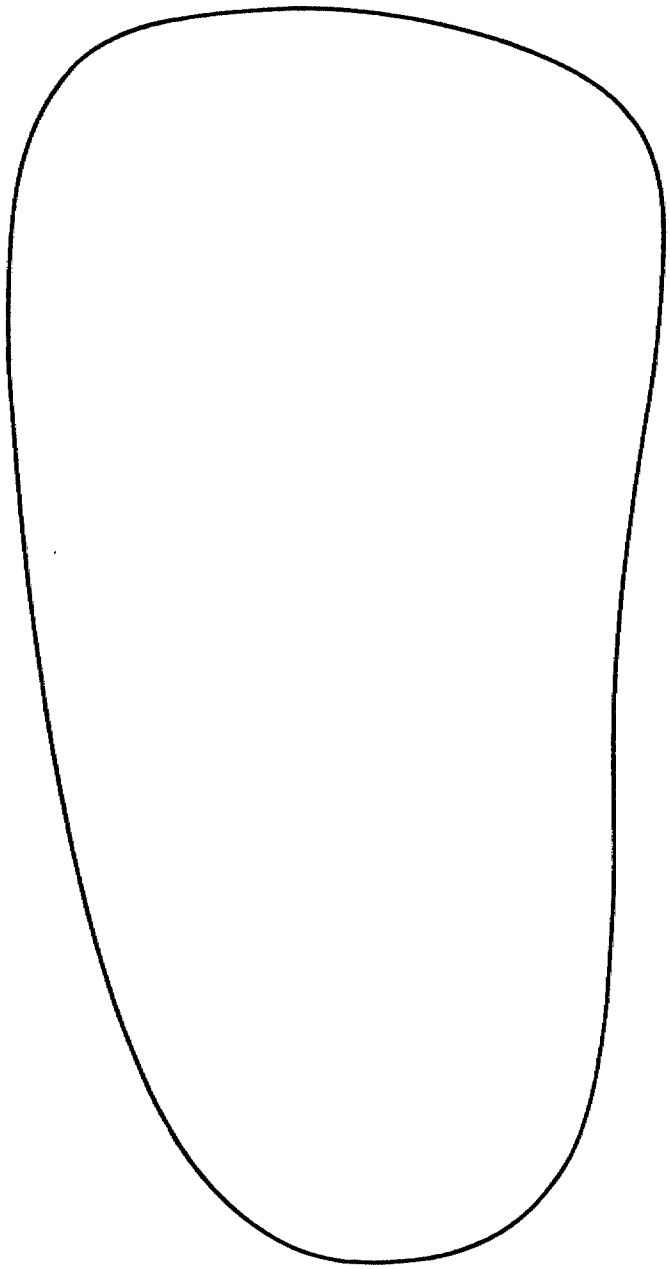


FIGURE 8. Arch support, plantar, male size 6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

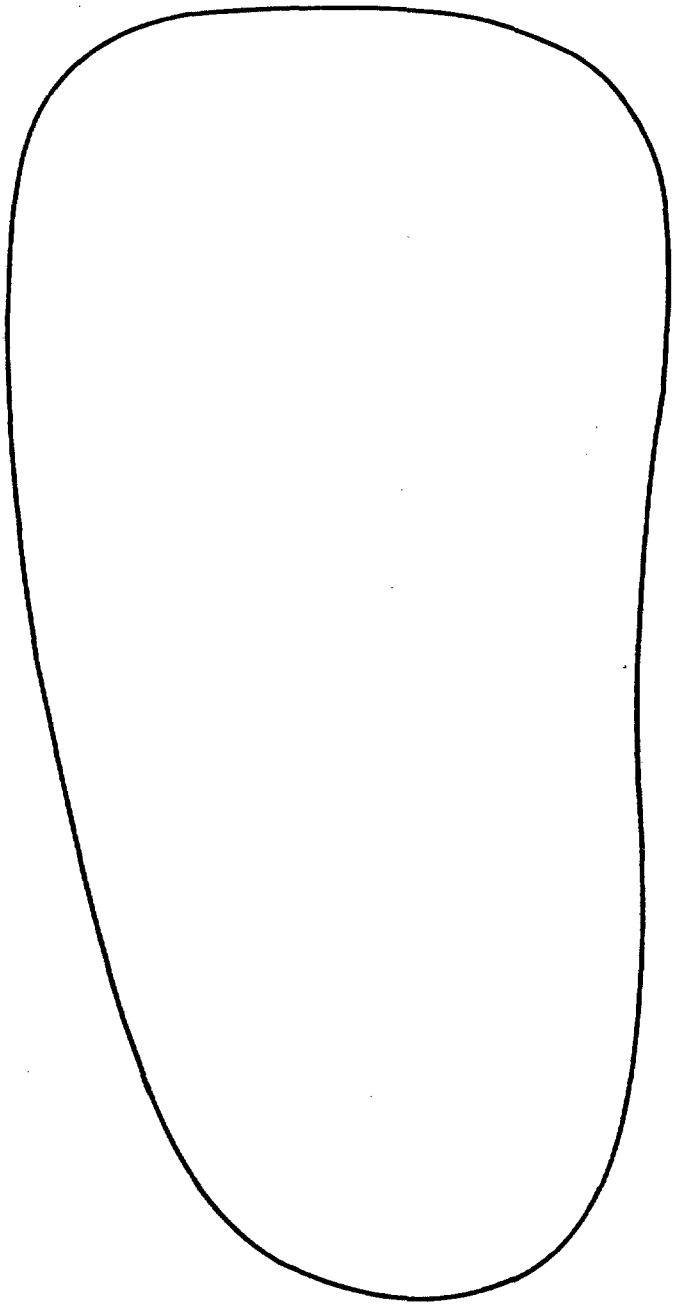


FIGURE 9. Arch support, plantar, male size 7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

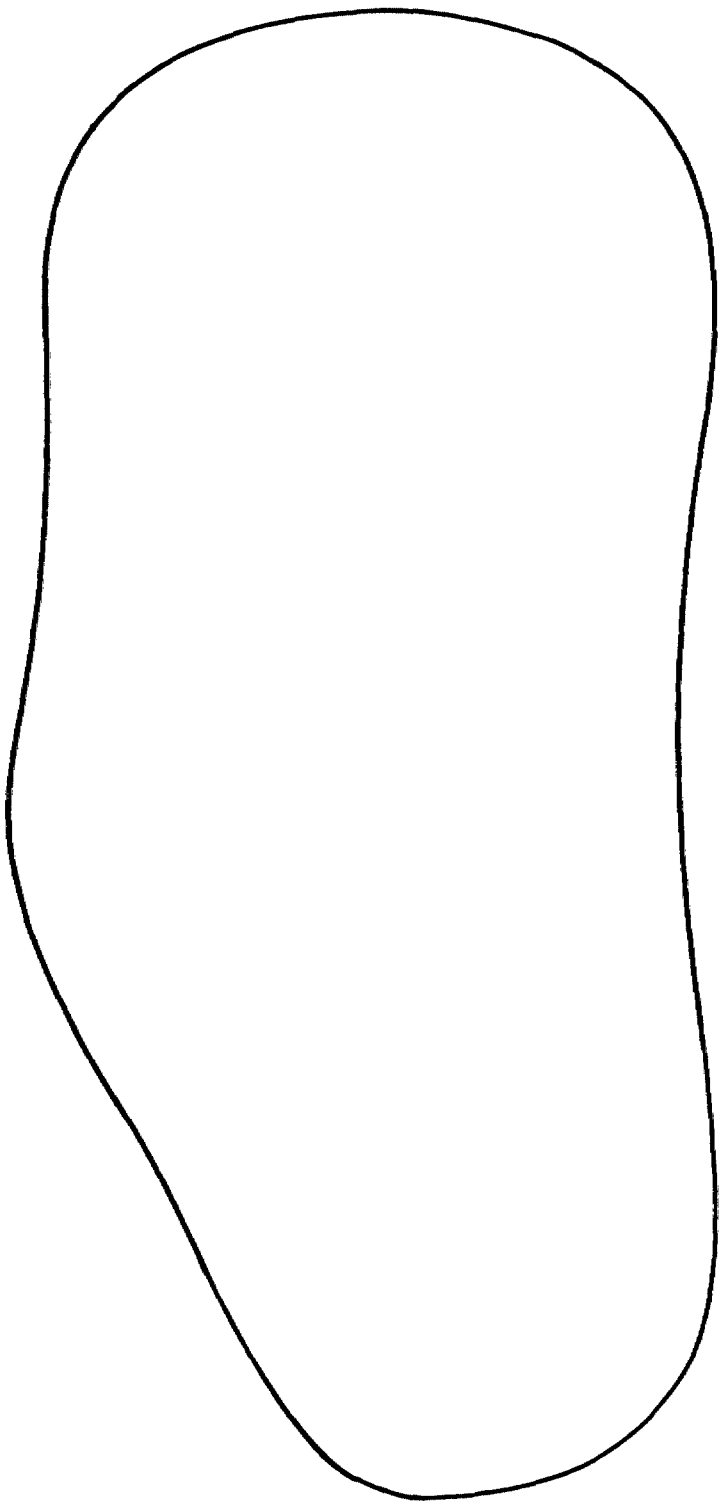


FIGURE 10. Arch support, combination female size 9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

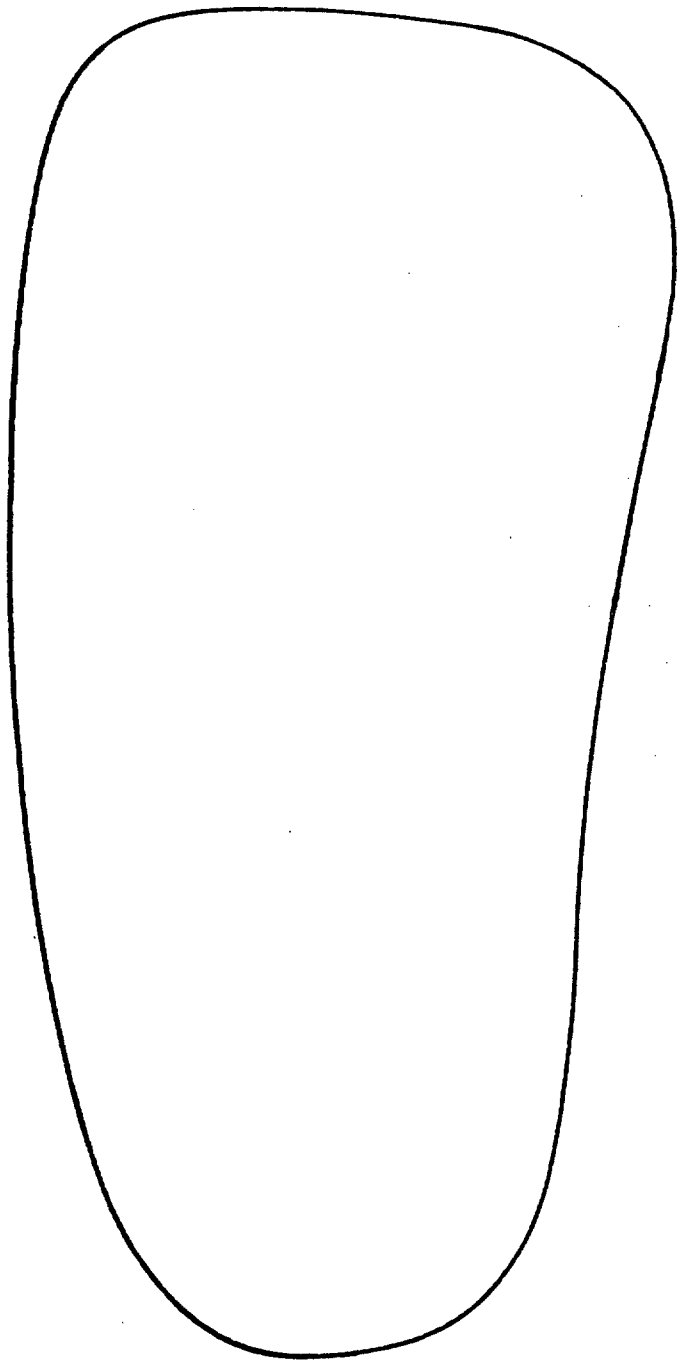


FIGURE 11. Arch support, combination, male size 6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

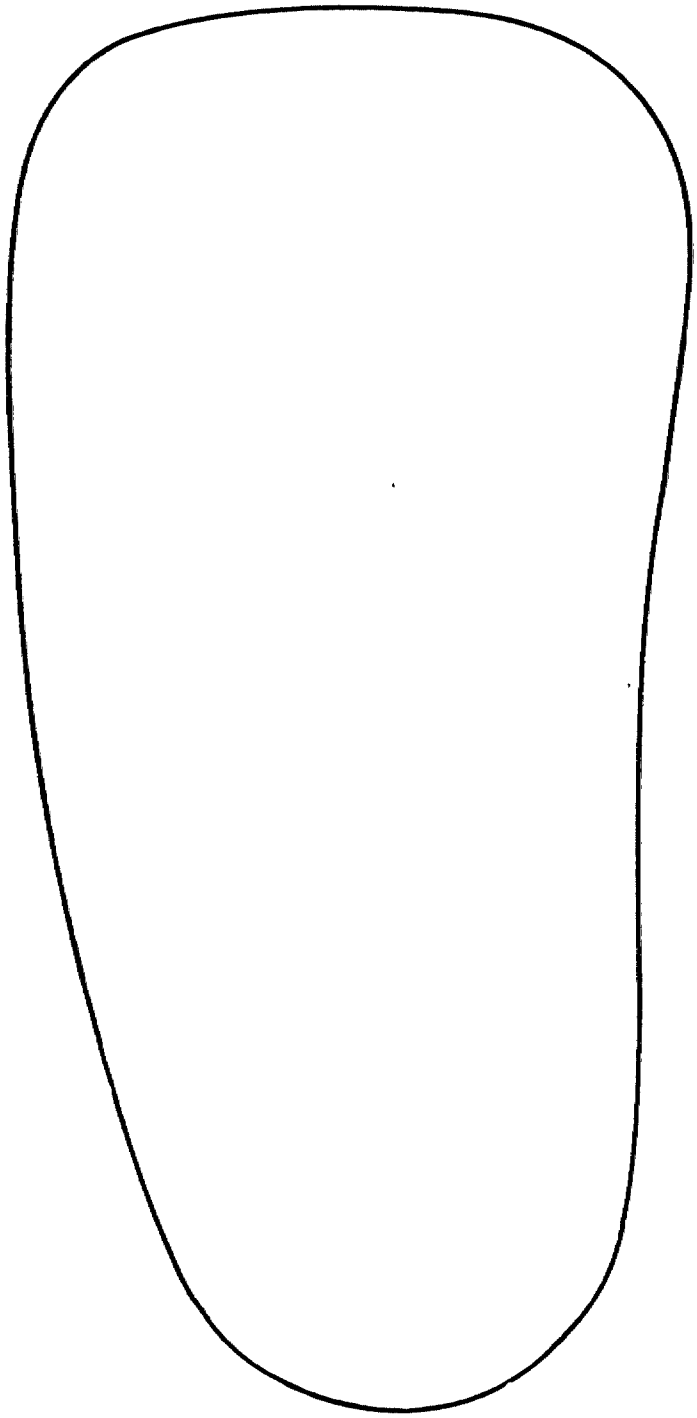


FIGURE 12. Arch support, combination, male size 7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

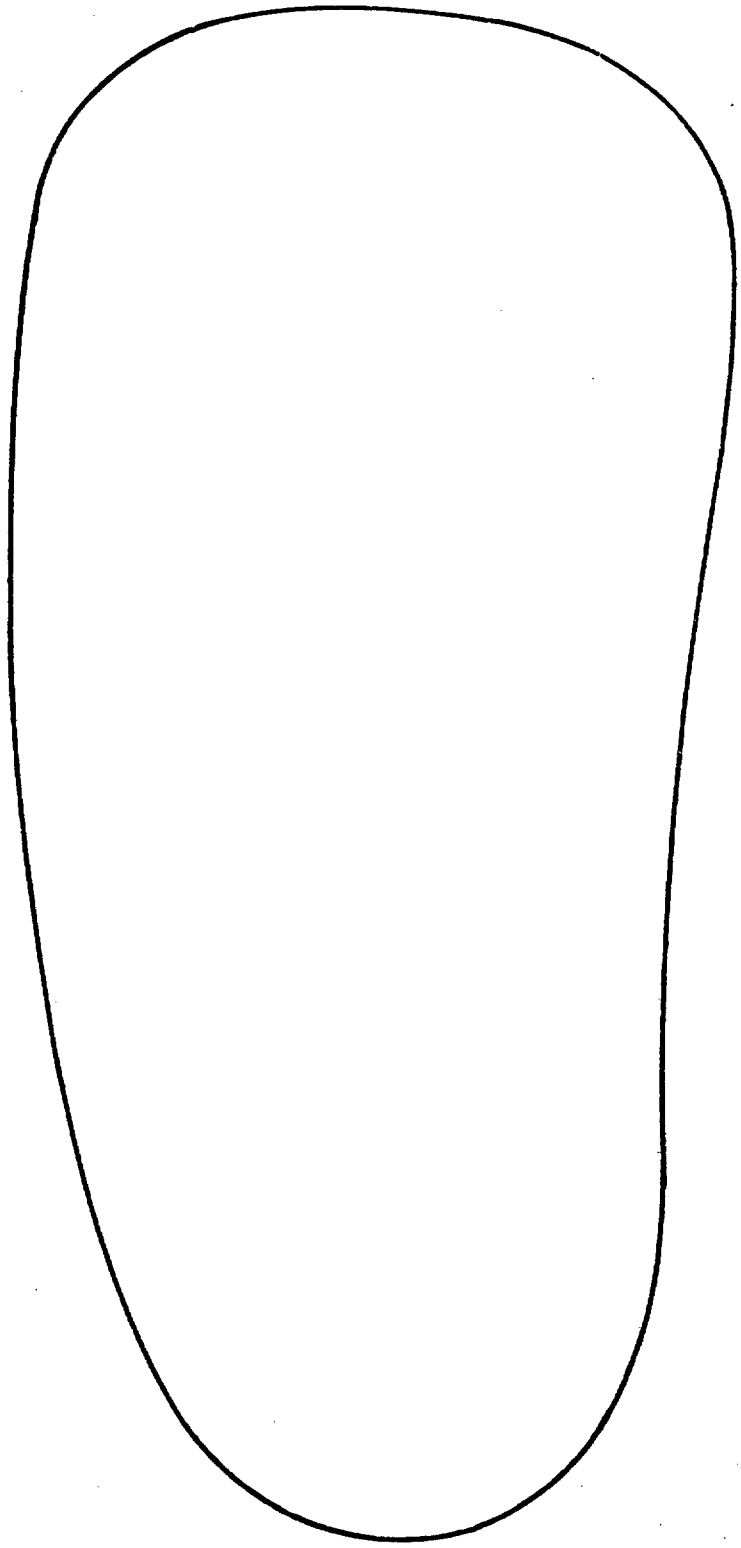


FIGURE 13. Arch support, combination, male size 10.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

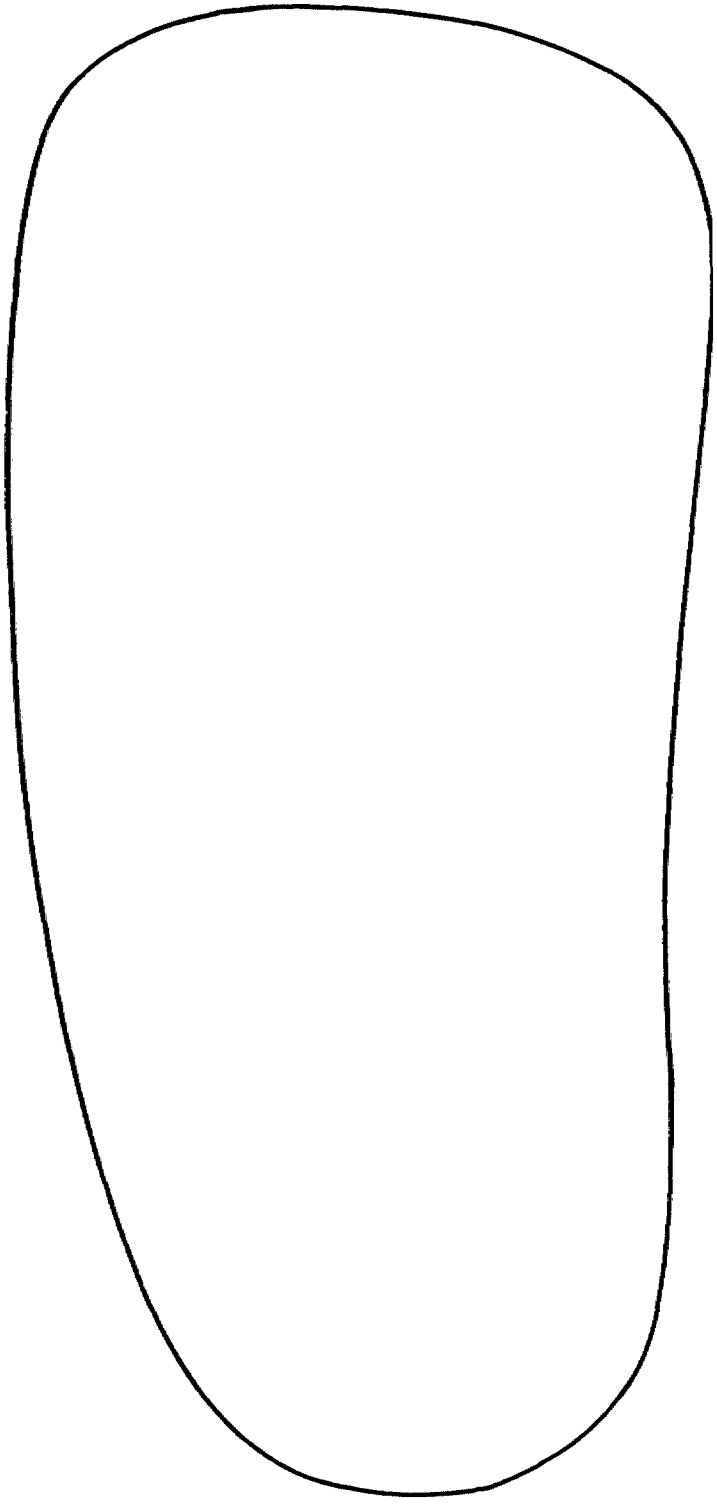


FIGURE 14. Arch support, combination, male size 11.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

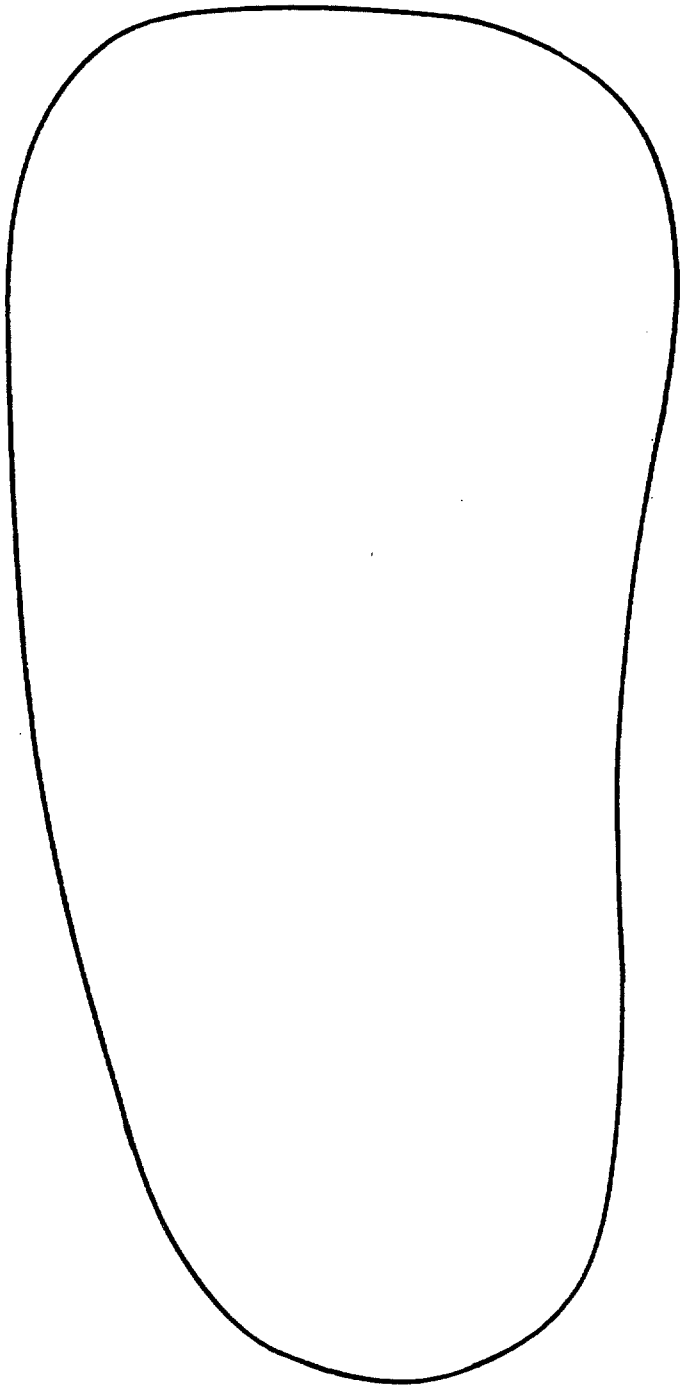


FIGURE 15. Arch support, combination, male size 8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

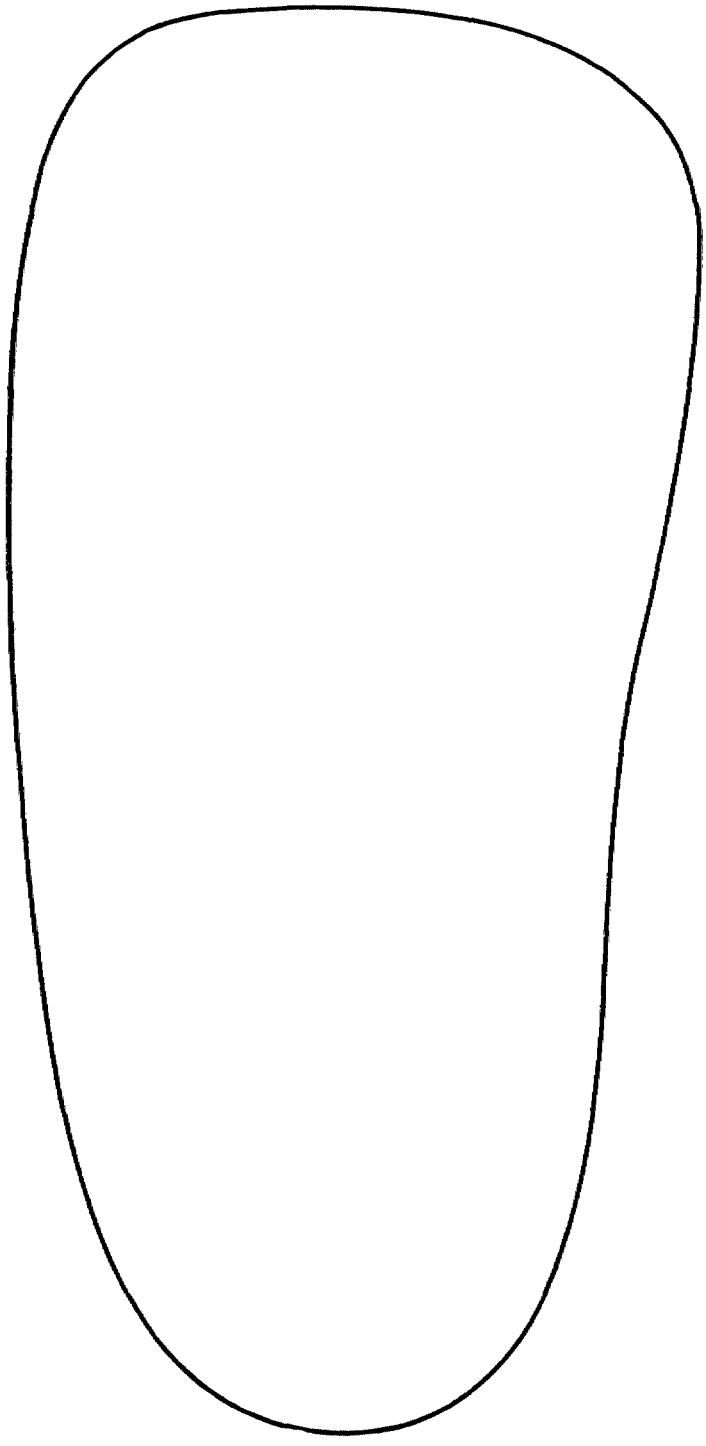


FIGURE 16. Arch support, combination, male size 9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

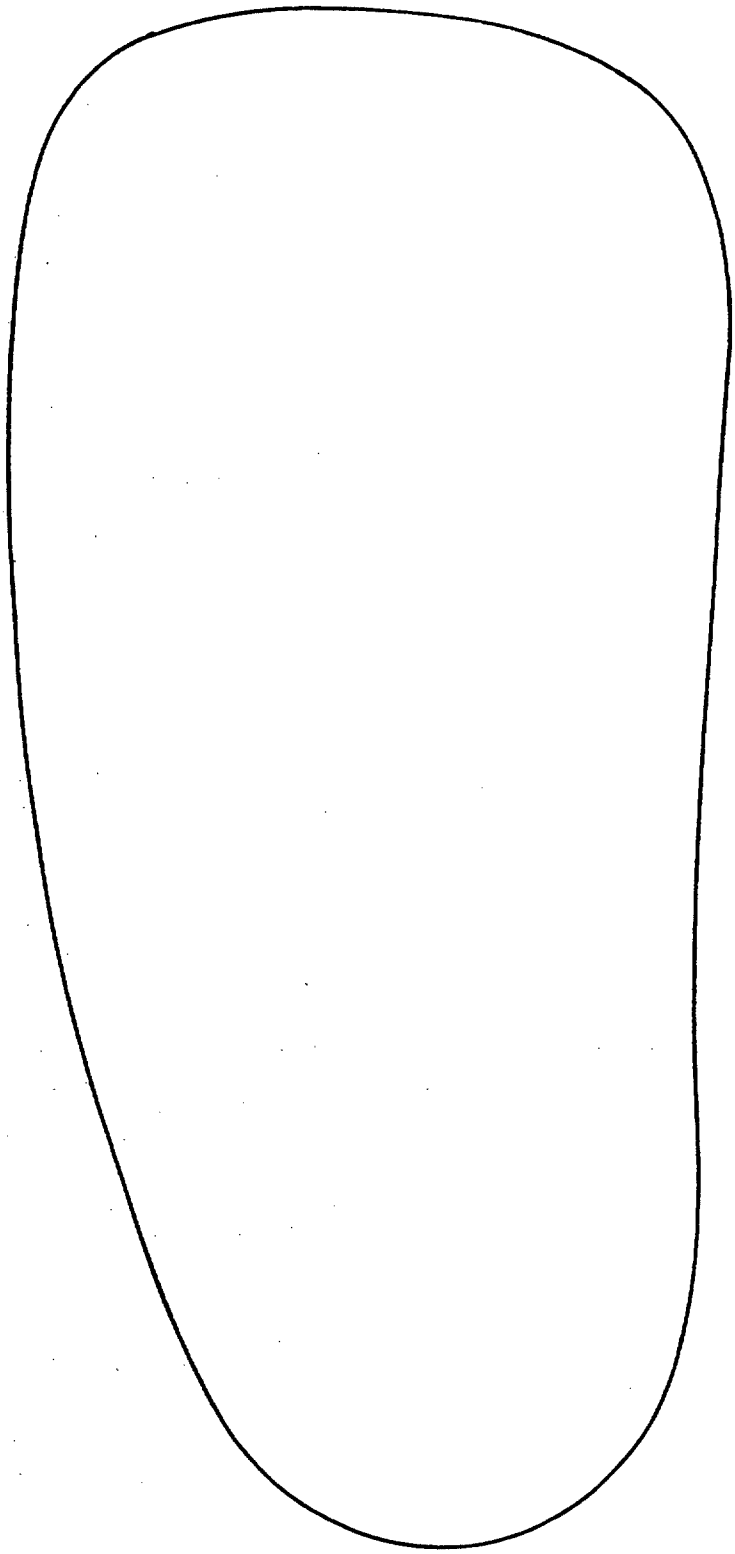


FIGURE 17. Arch support, combination, male size 12.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

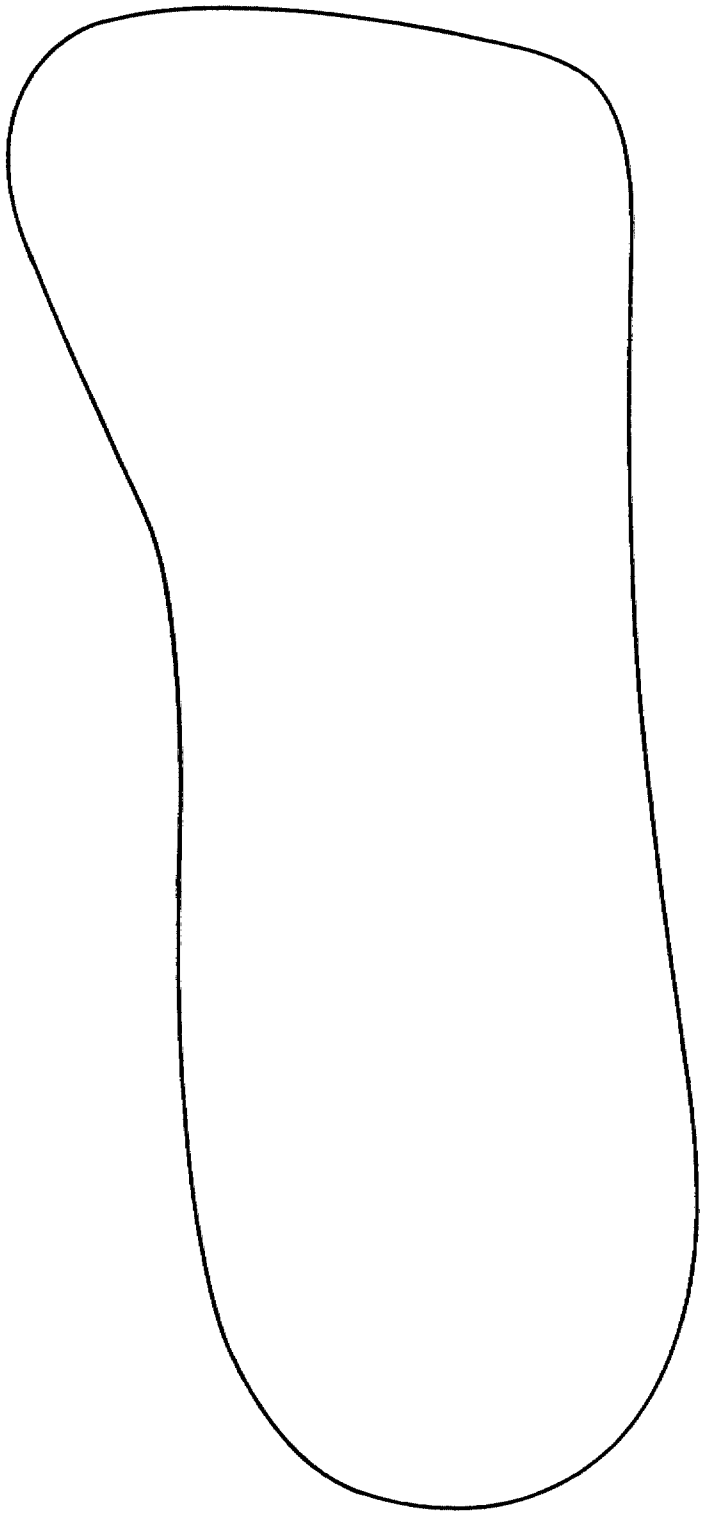


FIGURE 18. Arch support, metatarsal, male size 11.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

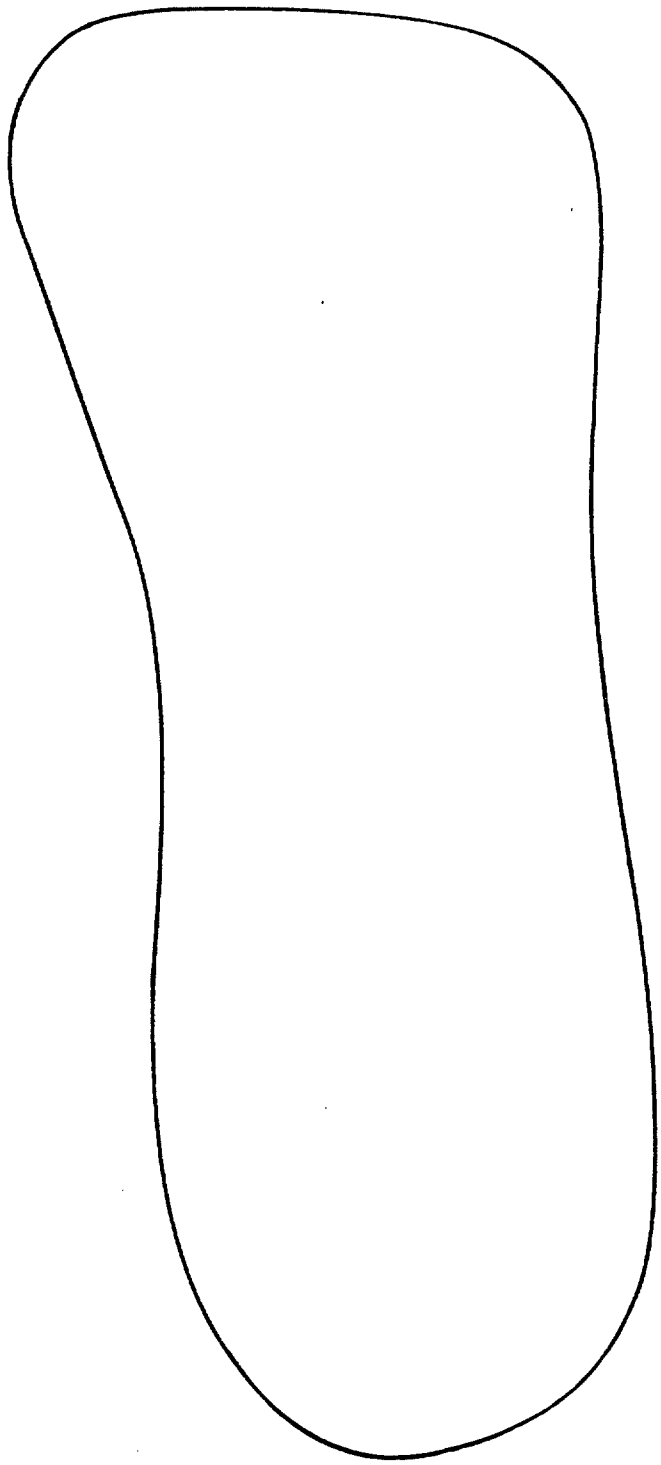


FIGURE 19. Arch support, metatarsal, male size 8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

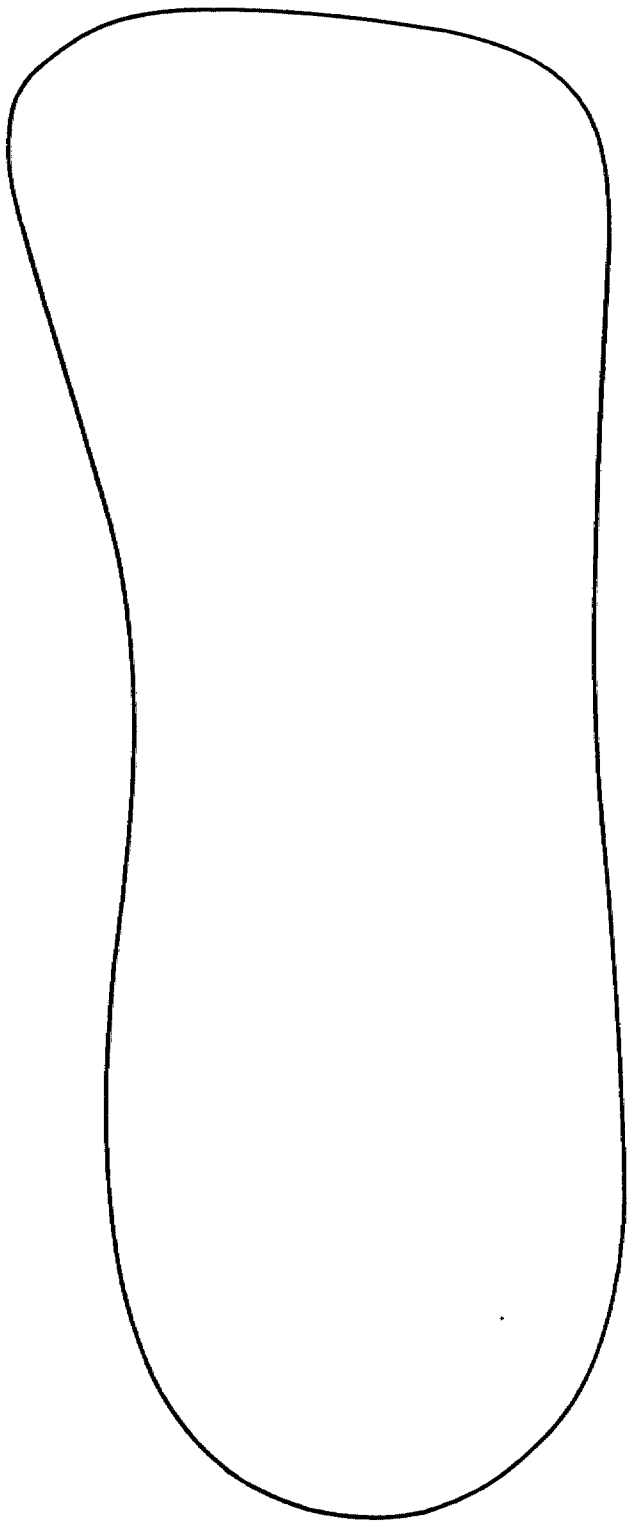


FIGURE 20. Arch support, metatarsal, male size 9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

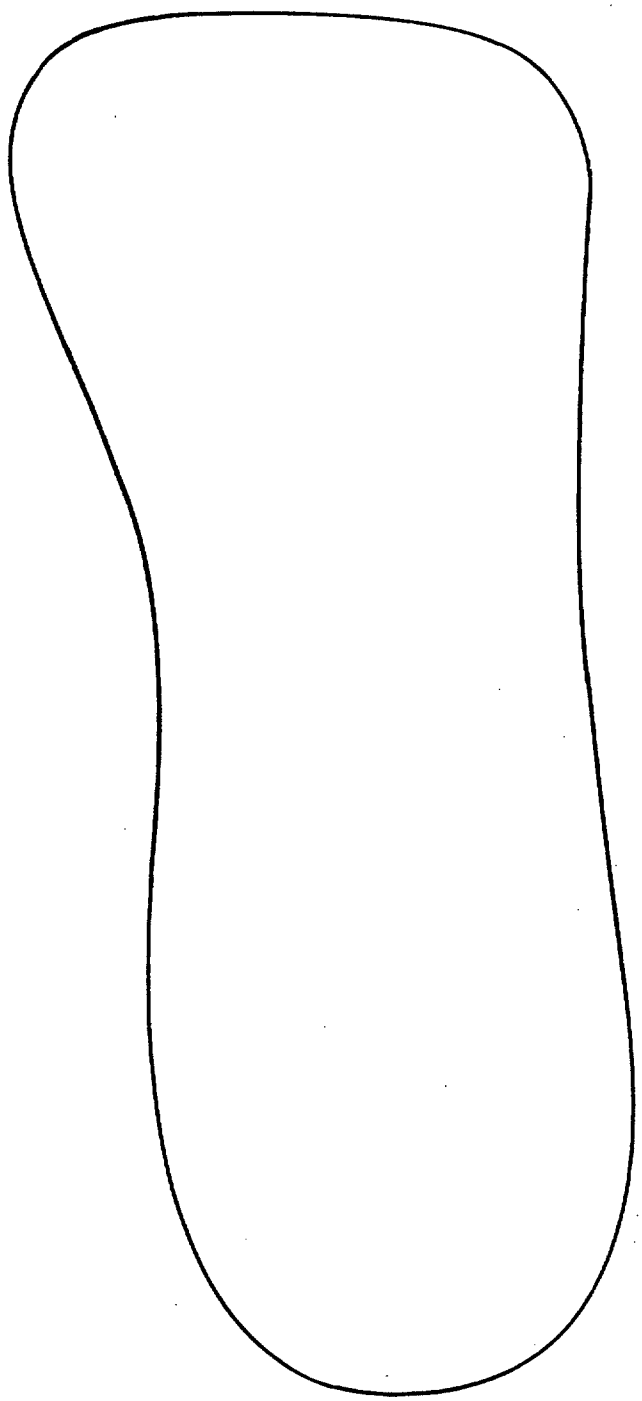


FIGURE 21. Arch support, metatarsal, male size 7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-A-3327A SUPPORT, ARCH

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 South 20th Street
Philadelphia, PA 19101

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY
DoD - 304

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:30Z
Check the source to verify that this is the current version before use.